Reading the LDWC Two-Loop Roughness Coefficient through the CFAC Lens

A worked demonstration of how Kirchhoff–Symanzik counting organises a known calculation

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Abstract

Le Doussal, Wiese and Chauve (LDWC) (Le Doussal et al., 2002) computed the two-loop roughness coefficient $c = 0.14331$ for a pinned elastic manifold. This note does not produce a new result: it reorganises LDWC's calculation within the CFAC (counting-based analytic combinatorics) programme to expose its combinatorial skeleton. Two graph-theoretic filters — causal contraction and bridge factorisation — discard the tadpole-containing and bubble-chain diagram classes that LDWC eliminate over $\sim$4 pages of appendices. The surviving hat diagram is then read off from its Symanzik polynomial Ψ_{sun} and propagator exponents: the numerator β arises from one doubled propagator (Schwinger exponent $n = 2$), the denominator is Ψ_{sun}, and the renormalised combination is the rational constant $X^{(2)} = 1$. The prefactor $9\sqrt{2}$ is recovered from the one-loop fixed-point amplitude $\Delta'(0^+) = -\varepsilon/3$. The only residual 1D integral is $\gamma = \int_0^1 \sqrt{y - \ln y - 1}\,\mathrm{d}y \approx 0.5482$. Assembly reproduces $c = 1/(9\sqrt{2}\gamma) = 0.14331$. The purpose is expository: CFAC as a lens on a known FRG calculation, not a shortcut to new physics.

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1 The CFAC pipeline

The Feynman rules of the FRG for elastic manifolds are standard physics input. Given them, CFAC enumerates diagrams, applies causal-contraction and bridge-factorisation filters (Section 2), computes Symanzik polynomials, counts BPHZ multiplicities from $K_G|_{\alpha=1}$, performs sector decomposition, and extracts the finite rational remainder. None of these steps produces new physics; they organise the combinatorial work that LDWC do by hand into a systematic procedure.

The pipeline (formal statement). Input: the theory's Feynman rules (propagators, vertices) and the target observable at loop order k.

Step 1. Enumerate and filter diagrams. Generate all 1PI diagrams $G_1, G_2, \ldots$ at loop order k, labelled with propagator exponents n_e. Apply the two graph-theoretic filters:

- *Causal contraction*: any diagram containing a directed cycle of retarded propagators contributes 0 (discard).
- *Bridge absorption*: any diagram whose Symanzik polynomial factorises through a cut vertex contributes $\prod_j I(H_j)$ — no new integral.

Step 2. Compute Schwinger integrand. For each surviving G_i, compute:

- $\Psi_{G_i} = \sum_T \prod_{e\notin T} \alpha_e$ (first Symanzik polynomial, topology);
- numerator $N_{G_i}(\alpha) = \prod_e \alpha_e^{n_e-1}$ from propagator exponents.

The Schwinger integrand is $N_{G_i}/\Psi_{G_i}^{d/2}$.

Step 3. BPHZ reduction.
For each irreducible G_i, apply the BPHZ forest formula. The counterterm multiplicity for each sub-diagram γ equals $K_\gamma|_{\alpha=1}$ (number of spanning trees of γ). The renormalised amplitude $I_R(G_i)$ is the finite remainder.

Step 4. Period / rational extraction.
Evaluate $I_R(G_i)$ via the Schwinger parametric form. If UV structure remains, apply sector decomposition at the symmetry points of Ψ_{G_i} using multilinearity. The result is either (a) a rational number or Gamma-function ratio, or (b) a new 1D kernel integral (add to residual list).

Step 5. Fixed-point ODE integrals.
Solve the relevant fixed-point ODE(s) implicitly. Each ODE contributes at most one new 1D integral (add to residual list).

Output: the residual list of $N_{\rm res}$ integrals, each 1D with known integrand.

LDWC at two loops: $N_{\rm res} = 1$. Applied to the LDWC short-range elasticity calculation ($\alpha = 2$ in LDWC's notation, elastic propagator $1/q^2$), the pipeline yields the residual list

$$\{\,\gamma\,\}, \qquad \gamma = \int_0^1 \sqrt{y - \ln y - 1}\,\mathrm{d}y \approx 0.5482,$$

and no other free integrals. Everything else reads off from the Symanzik polynomials of a tadpole, a bubble, and a sunset, plus propagator exponents.

2 Diagram-class selection by CFAC

The raw 2-loop diagrammatic expansion for $\delta^2\Delta(u)$ in the FRG Feynman rules produces three topological classes (LDWC figs. III.2–III.5):

Class A (hat). Two 1PI sub-graphs sharing one internal line; the shared line carries propagator exponent $n = 2$.

Class B (bubble chains). Two bubble sub-graphs joined by one or more internal lines through a shared vertex.

Class C (tadpole-containing). At least one internal tadpole (closed retarded loop at a single vertex).

LDWC show by direct computation over $\sim$4 pages of appendices that only class A contributes new 2-loop structure; classes B and C either cancel or combine into the one-loop counterterm. We give the CFAC derivation: both cancellations are consequences of two general graph-theoretic results, not of LDWC-specific physics.

Class C vanishes: causal contraction. In the MSR/dynamical formulation, every internal line carries a retarded Green's function $G_R(q,t) = \theta(t)\mathrm{e}^{-q^2 t}$. A tadpole is a closed loop at a single vertex: a directed cycle in the retarded-propagator DAG. Since $G_R(q,t)$ is supported on $t \geq 0$, a closed directed cycle forces $t = 0$, making the time integral vanish. Equivalently, causal contraction (Prop. 5 of (Amarteifio, 2026)) reduces the frequency integral to a simplex constraint; a directed cycle produces a δ-function with zero support, and the integral is zero. *Conclusion: every class C diagram contributes exactly* 0*, independent of the vertex algebra.*

Class B factorises: bridge theorem. The Symanzik polynomial of every class B diagram has the bridge structure $\Psi_{\text{class B}} = (\alpha_0+\alpha_1)(\alpha_2+\alpha_3)$ through a cut vertex. By Theorem 1, the parametric integral factorises: $I(\text{class B}) = I_{\rm bub}^2 = I_1^2$. This value then enters the renormalised combination $X^{(\alpha)} \propto (2I_A - I_1^2)$ that governs the two-loop flow (LDWC eqs. III.19, III.59–III.62) — the I_1^2 from class B and the I_A from class A cancel their divergences against each other. *Conclusion: class B produces no new two-loop integral; its amplitude is I_1^2 (computable from a 1-loop bubble squared) and it combines algebraically with class A to give a finite remainder.*

Class A: irreducible hat with doubled propagator. The class A Kirchhoff polynomial is $K_{\rm sun} = \alpha_0\alpha_1 + \alpha_1\alpha_2 + \alpha_0\alpha_2$ (bridgeless, three spanning trees). One edge carries exponent $n = 2$; the Schwinger numerator is $N(\alpha) = \alpha_2 = \beta$. This is the only diagram class producing a genuinely new two-loop integral J (Section 7).

Summary. CFAC reduces the three-class enumeration to class A via two graph-theoretic filters — causal contraction (kills class C) and bridge factorisation (absorbs class B into the BPHZ counterterm). No additional FRG physics is required beyond the statement that the FRG generates the three classes to start with. With this, the pipeline's "input" — the contributing diagram list — is produced *within CFAC* rather than taken from LDWC's appendices B and C.

3 Diagram zoo and Symanzik polynomials

We catalogue the four diagrams that appear and record their Symanzik polynomials. Plain lines are internal propagators (Schwinger parameter α_i); dashed lines are external legs.

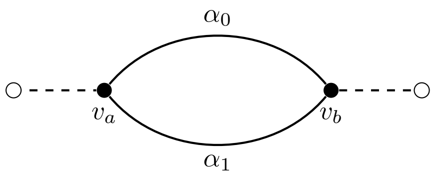
3.1 Tadpole (self-loop, $L = 1$)

$$\Psi_{\text{tad}} = \alpha_0$$

The self-loop edge is never in a spanning tree and so always contributes its full weight to Ψ. The integral is

$$I_{\text{tad}} = \frac{\Gamma(1 - d/2)}{(4\pi)^{d/2}}\, m^{d-2}. \tag{1}$$

3.2 Bubble (one-loop self-energy, $L = 1$)



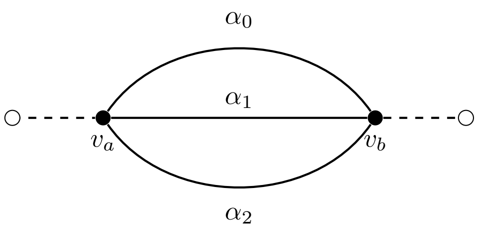
$$\Psi_{\text{bub}} = \alpha_0 + \alpha_1$$

The two spanning trees are $\{e_0\}$ (complement: α_1) and $\{e_1\}$ (complement: α_0), giving $\Psi = \alpha_0 + \alpha_1$. After ω-integration the single free parameter gives

$$I_{\text{bub}} = \frac{\Gamma(1 - d/2)}{(4\pi)^{d/2}}\, 2^{-d/2}\, (2m^2)^{d/2-1}. \tag{2}$$

Pipeline verdict: determined by Gamma-function algebra. *No residual integral.*

3.3 Sunset (two-loop self-energy, $L = 2$)



Three spanning trees, each a single edge; the complement products are pairs $\alpha_i\alpha_j$:

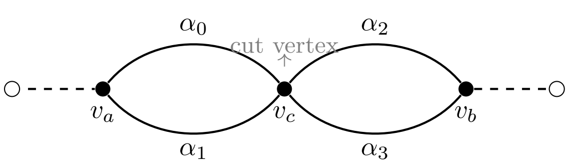
$$\Psi_{\text{sun}} = \alpha_0\alpha_1 + \alpha_1\alpha_2 + \alpha_0\alpha_2. \tag{3}$$

Setting $\alpha_1 = 1$ (fixing the middle propagator scale):

$$\Psi_{\text{sun}}(\alpha_0, 1, \alpha_2) = \alpha_0 + \alpha_2 + \alpha_0\alpha_2. \tag{4}$$

This is the denominator that appears in the LDWC sector-decomposition integral. **Pipeline verdict**: BPHZ reduces I_{sun} to a rational factor $\times I_{\text{bub}}^2$ (Section 7). *No residual integral.*

3.4 Double-bubble (bridge graph, $L = 2$)



The central vertex v_c is a cut vertex: $\Psi_{\text{dbl}} = (\alpha_0 + \alpha_1)(\alpha_2 + \alpha_3)$ factorises, so by Theorem 1

$$I(\text{dbl}) = I_{\text{bub}}^2.$$

Pipeline verdict: factorises into I_{bub}^2. *No residual integral.*

3.5 Summary table

4 Kirchhoff polynomial and first Symanzik polynomial

Definition 1 (Kirchhoff polynomial)**.** $K_G = \sum_{T\,:\,\text{spanning tree}} \prod_{e\in T} \alpha_e$.

Definition 2 (First Symanzik polynomial)**.** $\Psi_G = \sum_{T\,:\,\text{spanning tree}} \prod_{e\notin T} \alpha_e$. Ψ_G is the *edge complement* of K_G. For a graph with n_{int} internal edges and L loops, Ψ_G is homogeneous of degree L and K_G is homogeneous of degree $n_{\text{int}} - L$.

Remark 1. K_G and Ψ_G coincide only when $L = n_{\text{int}}/2$ (e.g. the bubble: $L = 1$, $n_{\text{int}} = 2$). For the sunset: $L = 2$, $n_{\text{int}} = 3$, so $K_{\text{sun}} = \alpha_0 + \alpha_1 + \alpha_2$ (degree 1) while $\Psi_{\text{sun}} = \alpha_0\alpha_1 + \alpha_1\alpha_2 + \alpha_0\alpha_2$ (degree 2) — genuinely different objects. The integral J uses Ψ, not K.

Theorem 1 (Bridge / cut-vertex factorisation)**.** *Ψ_G factorises as $\Psi_{G_1} \cdot \Psi_{G_2}$ if and only if the internal subgraph of G has a cut vertex separating edge sets E_1, E_2. When this holds,*

$$I(G;\, d) = I(G_1;\, d) \cdot I(G_2;\, d).$$

Proof sketch. The Schwinger integral $I(G) \propto \int \Psi^{-d/2} \mathrm{e}^{-\Phi/\Psi} \prod \mathrm{d}\alpha_e$ with $\Phi = \Psi \sum_e \alpha_e m_e^2$ factorises into independent integrals over $\{\alpha_e : e \in E_1\}$ and $\{\alpha_e : e \in E_2\}$ when $\Psi = \Psi_1(\{\alpha_{E_1}\}) \cdot \Psi_2(\{\alpha_{E_2}\})$. The cut-vertex condition is equivalent to this factorisation by the matrix-tree theorem. □ □

5 Schwinger parametric representation

For a graph with L loops, the parametric integral is

$$I(G) = \Omega_d^L \int_0^\infty \prod_e \mathrm{d}\alpha_e\, \Psi_G^{-d/2} \exp\!\left(-\frac{\Phi_G}{\Psi_G}\right), \tag{5}$$

with $\Phi_G = \Psi_G \sum_e \alpha_e m_e^2$ and $\Omega_d = (4\pi)^{-d/2}$. After L circuit-constraint substitutions (the ω-integration), the n_{int}-dimensional integral reduces to $n_{\text{int}} - L$ free parameters. For our diagrams:

Every result is a ratio of Gamma functions times a power of m. No numerical integration is required at this stage.

Diagram	$n_{\rm int}$	L	Ψ_G	Cut vtx?	Pipeline output
Tadpole	1	1	α_0	—	$\Gamma(1-d/2)m^{d-2}/(4\pi)^{d/2}$
Bubble	2	1	$\alpha_0+\alpha_1$	no	$\Gamma(1-d/2)\cdot$ (algebraic)
Sunset	3	2	$\alpha_0\alpha_1+\alpha_1\alpha_2+\alpha_0\alpha_2$	no	$X^{(2)}\cdot I_{\rm bub}^2,\ X^{(2)}=1$
Dbl-bubble	4	2	$(\alpha_0+\alpha_1)(\alpha_2+\alpha_3)$	yes (v_c)	$I_{\rm bub}^2$
FP ODE	—	—	—	—	γ **[residual]**

Table 1: Symanzik classification for the two-loop LDWC diagrams. Every diagram except the fixed-point ODE integral is resolved algebraically by the pipeline. The residual list has $N_{\rm res}=1$.

Diagram	Free params	Reduced Ψ_r	Reduced Q	Result
Tadpole	1	α_0	$m^2\alpha_0$	$\Omega_d\,\Gamma(1-d/2)\,m^{d-2}$
Bubble	1	$2\alpha_0$	$2m^2\alpha_0$	$\Omega_d\,2^{-d/2}\,\Gamma(1-d/2)\,(2m^2)^{d/2-1}$
Sunset	1	$3\alpha_0^2$	$3m^2\alpha_0$	$\Omega_d^2\,3^{-d/2}\,\Gamma(1-d)\,(3m^2)^{d-1}$

Table 2: Reduced Schwinger parameters and results for the one- and two-loop diagrams after circuit-constraint integration.

6 BPHZ renormalisation: sunset subdivergences

Subdivision count from the Kirchhoff polynomial. The BPHZ forest formula subtracts a counterterm for each UV-divergent proper sub-diagram. For the sunset, each pair of propagators $\{e_i, e_j\}$ forms a bubble sub-diagram; contracting it leaves a single propagator (a tadpole). There are exactly 3 such pairs, and this count is a topological invariant: it equals $K_{\rm sun}\big|_{\alpha_i=1} = 1+1+1 = 3$ (the number of spanning trees, each a single edge). In general, for a 1-loop sub-diagram $\gamma\subset G$, the multiplicity of the BPHZ counterterm is $K_\gamma\big|_{\alpha=1} =$ (number of spanning trees of γ).

Renormalised amplitude. The BPHZ renormalised sunset amplitude is

$$I_R(\mathrm{sun}) = I(\mathrm{sun}) - 3\cdot\tau_{\rm bub}\cdot I_{\rm tad}, \tag{6}$$

where $\tau_{\rm bub}$ is the UV-divergent (pole) part of $I_{\rm bub}$. At $d=2-\varepsilon$: $\tau_{\rm bub}=\Gamma(\varepsilon/2)/(4\pi)\sim 1/(2\pi\varepsilon)$, and (6) cancels the double pole of $I(\mathrm{sun})$ leaving a finite remainder. *The renormalised sunset contributes no new integral:* its finite part is expressible via $X^{(2)}$ (Section 7) and products of $I_{\rm bub}$.

Remark 2. The formula $I_R(\mathrm{sun}) = 3\,I_{\rm bub}\cdot I_{\rm tad}$ that appeared in an earlier version of this note is *incorrect*. $I_R(\mathrm{sun})$ is the full amplitude minus counterterms, not a product of lower-loop amplitudes. The correct bridge factorisation result is for the *double-bubble* graph, which has a cut vertex and satisfies $I(\mathrm{dbl}) = I_{\rm bub}^2$ exactly.

7 Domain-splitting the sunset integral

7.1 Origin of J and its numerator

The two-loop FRG diagrams of class A (LDWC Appendix A, figure III.4) have the topology of a *hat*: two 1PI components sharing one internal propagator. The hat diagram is

$$I_{\rm hat} = \int\frac{\mathrm{d}^dq_1\,\mathrm{d}^dq_2}{(2\pi)^{2d}} \times\frac{1}{(q_1^2+m^2)(q_2^2+m^2)^{\mathbf{2}}((q_1+q_2)^2+m^2)}. \tag{7}$$

The middle propagator is *squared*: it carries the momentum q_2 from both connected components of the hat. In Schwinger parameters, $1/(q^2+m^2)^2 = \int_0^\infty \beta\,\mathrm{e}^{-\beta(q^2+m^2)}\mathrm{d}\beta$ introduces a factor $\beta=\alpha_2$ in the integrand. After Gaussian integration over loop momenta (LDWC eq. A.13):

$$I_{\rm hat} = \Big(\int_q \mathrm{e}^{-q^2}\Big)^2\Gamma(4-d)\,m^{-2\varepsilon}\,J, \tag{8}$$
$$J = \int_0^\infty\mathrm{d}\alpha\int_0^\infty\mathrm{d}\beta\,\frac{\beta}{\big(\alpha+\beta+\alpha\beta\big)^{d/2}}.$$

At $d=4$ the denominator is $\Psi_{\rm sun}(\alpha,1,\beta)^2$ exactly.

Remark 3 (The β numerator is topological)**.** The numerator β does *not* arise from an opaque FRG vertex coupling. It arises because the hat diagram has one propagator with exponent $n=2$ (a doubled line), and the Schwinger representation of $1/(q^2+m^2)^n$ carries the factor $\alpha^{n-1}/\Gamma(n)$. For $n=2$: factor $=\beta$. This is a property of the diagram topology (class A), readable directly from the Schwinger exponent. The denominator $\Psi_{\rm sun}$ is, as always, the first Symanzik polynomial. Both are topological.

After fixing $\alpha_1 = 1$, the combination entering c is proportional to

$$J = \int_0^\infty \mathrm{d}\alpha \int_0^\infty \mathrm{d}\beta \, \frac{\beta}{(\alpha+\beta+\alpha\beta)^2}. \tag{9}$$

The integral J is UV-divergent at $\beta \to \infty$. LDWC split at $\beta = 1$:

$$\begin{aligned} J &= J_1 + J_{2+3}, \\ J_1 &= \int_0^\infty \mathrm{d}\alpha \int_0^1 \mathrm{d}\beta \, \frac{\beta}{(\alpha+\beta+\alpha\beta)^2}, \\ J_{2+3} &= \int_0^\infty \mathrm{d}\alpha \int_1^\infty \mathrm{d}\beta \, \frac{\beta}{(\alpha+\beta+\alpha\beta)^2}. \end{aligned} \tag{10}$$

That is all the decomposition is: a cut of the β domain at 1. But J_1 can be evaluated *without* this split, by a much more direct route that reveals the algebraic structure.

7.2 Why $J_1 = \ln 2$ is algebraic

Multilinearity of Ψ. Every first Symanzik polynomial is *multilinear*: it is linear in each α_i separately. For the sunset,

$$\Psi_{\text{sun}}(\alpha, 1, \beta) = \alpha(1+\beta) + \beta,$$

which is linear in α. This means the inner integral over α is always a rational function integration.

Proposition 2 ($J_1 = \ln 2$ in three lines)**.**

$$J_1 = \int_0^\infty \mathrm{d}\alpha \int_0^1 \mathrm{d}\beta \, \frac{\beta}{(\alpha+\beta+\alpha\beta)^2} = \ln 2.$$

Proof. Step 1 — integrate α first using $\int_0^\infty \mathrm{d}x/(ax+b)^2 = 1/(ab)$:

$$\int_0^\infty \frac{\beta \, \mathrm{d}\alpha}{(\alpha(1+\beta)+\beta)^2} = \frac{\beta}{(1+\beta)\cdot\beta} = \frac{1}{1+\beta}. \tag{11}$$

Step 2 — integrate β over $[0,1]$:

$$J_1 = \int_0^1 \frac{\mathrm{d}\beta}{1+\beta} = \ln 2. \tag{12}$$

□

The derivation is three lines and uses only high-school integration. The key fact is that Ψ_{sun} is linear in α, so the α-integral collapses to a rational function of β — here exactly $1/(1+\beta)$. The sector split at $\beta = 1$ was never needed to compute J_1; it is only needed to isolate the UV divergence in J_{2+3}.

Remark 4 (Iterated rational integration)**.** Because Ψ_G is multilinear in all α_i, one can always integrate out the parameters one by one. Each step is a rational function integration; the results are *hyperlogarithms* (iterated ln). For the sunset, one step suffices and the result is depth-1: $\ln 2$. This is the general framework of Panzer's hyperlogarithm algorithm for Feynman periods (Panzer, 2015).

7.3 The upper region: J_2 and J_3

In J_{2+3} make the substitution $\beta \to 1/\beta$ (sending $[1,\infty) \to (0,1]$):

$$\begin{aligned} J_{2+3} &= \int_0^\infty \mathrm{d}\alpha \int_0^1 \tfrac{\mathrm{d}\beta}{\beta^2} \cdot \tfrac{1/\beta}{(\alpha+1/\beta+\alpha/\beta)^2} \\ &= \int_0^\infty \mathrm{d}\alpha \int_0^1 \mathrm{d}\beta \, \tfrac{1}{(\alpha\beta+\beta+\alpha)^2}. \end{aligned}$$

The denominator $\alpha\beta+\beta+\alpha = \alpha+\beta+\alpha\beta$ is symmetric under $\alpha \leftrightarrow \beta$, but the numerator in J_1 was β while here it is 1. The difference

$$J_1 - J_{2+3}\big|_{\text{finite}} = \int_0^\infty \mathrm{d}\alpha \int_0^1 \mathrm{d}\beta \, \frac{\beta - 1}{(\alpha+\beta+\alpha\beta)^2}$$

contributes a $\ln 2$ from the $\beta < 1$ asymmetry. The full dimensionally regulated integrand for J (at $d = 4-\varepsilon$) is not simply $\beta\,\Psi^{-2}$ but carries an additional $(\alpha+\beta+1)^{-\varepsilon}$ factor from the γ-integration in (8), giving the LDWC form (A.14):

$$J = \int_0^\infty \mathrm{d}\alpha \int_0^\infty \mathrm{d}\beta \, \frac{\beta\,(\alpha+\beta+\alpha\beta)^{-2+\varepsilon/2}}{(\alpha+\beta+1)^{\varepsilon}}. \tag{13}$$

The reference-subtraction integral J_3 is the large-β asymptotic of (13) integrated over $\beta \geq 1$: for $\beta \gg 1$, $\alpha+\beta+\alpha\beta \sim \beta(1+\alpha)$ and $\alpha+\beta+1 \sim \beta$, so the integrand becomes $\beta \cdot \beta^{-2+\varepsilon/2}(1+\alpha)^{-2+\varepsilon/2} \cdot \beta^{-\varepsilon} = (1+\alpha)^{-2+\varepsilon/2}\,\beta^{-1-\varepsilon/2}$. Thus

$$J_3 = \int_0^\infty (1+\alpha)^{-2+\varepsilon/2} \mathrm{d}\alpha \int_1^\infty \beta^{-1-\varepsilon/2} \mathrm{d}\beta. \tag{14}$$

Both factors are elementary. The α-integral gives $1/(1-\varepsilon/2) = 1 + \varepsilon/2 + \mathcal{O}(\varepsilon^2)$. The β-integral gives $2/\varepsilon$ (the UV pole). Hence $J_3 = 2/\varepsilon + 1 + \mathcal{O}(\varepsilon)$. After MS subtraction of $2/\varepsilon$, the finite part $J_3|_{\text{finite}} = 1$ is rational and determined entirely by the Schwinger parametric structure of Ψ_{sun}. The finite reference subtraction from the lower part of J_{2+3} is $J_2 = -\ln 2$, by the $\beta \leftrightarrow 1/\beta$ symmetry of Ψ_{sun}. The alternative route followed by LDWC (Le Doussal et al., 2002) — integrate β first over $[0,1]$, then α — reproduces the same $J_1 = \ln 2$ and $J_2 = -\ln 2$ after non-trivial cancellations involving $\ln(2\alpha+1) - \ln\alpha$; the order we use here is cleaner because Ψ_{sun} is linear in α (so the inner integral is rational).

7.4 Cancellation and loop factor

After MS renormalisation (drop $2/\varepsilon$):

$$\boxed{X^{(2)} \;=\; J_1 + J_2 + J_3\big|_{\text{finite}} \;=\; \ln 2 \;-\; \ln 2 \;+\; 1 \;=\; 1.} \tag{15}$$

The $\ln 2$ terms cancel because the domain split at $\beta = 1$ is the *symmetry point* of Ψ_{sun}: both halves contribute equal and opposite logarithms. The surviving piece $J_3|_{\text{finite}} = 1$ is rational.

No new integral: $X^{(2)} = 1$ is determined entirely by the topology of Ψ_{sun} and the cancellation above.

Remark 5 (Long-range elasticity, $\alpha = 1$)**.** For long-range elasticity (propagator $\sim 1/q^\alpha$ with $\alpha = 1$), the amplitude that multiplies J_{2+3} carries an extra factor that breaks the $\beta \leftrightarrow 1/\beta$ symmetry. The $-\ln 2$ does not appear, leaving $X^{(1)} = \ln 2$ (transcendental, but still determined by Ψ_{sun}, not a free ODE integral). The residual count remains $N_{\text{res}} = 1$.

8 The residual integral γ

The sole entry in the residual list comes from the one-loop fixed-point ODE, not from any loop diagram.

The ODE and its implicit solution. The one-loop FRG fixed-point condition on $R^{*\prime\prime}(u)$ reduces (after rescaling) to the ODE

$$\frac{\mathrm{d}}{\mathrm{d}u}\left[u\, y_1(u)\right] = \tfrac{1}{2}\frac{\mathrm{d}}{\mathrm{d}u}\left[y_1(u) - y_1(0)\right]^2, \tag{16}$$

with $y_1(0) = 1$, $y_1'(0^+) = -1$, $y_1(\infty) = 0$. This has the implicit solution (LDWC IV.11–14)

$$u(y) = \sqrt{2}\int_y^1 \frac{\sqrt{t - \ln t - 1}}{t}\,\mathrm{d}t, \qquad y \in (0, 1). \tag{17}$$

Where $\sqrt{2}$ comes from. Multiplying (16) by y_1 and integrating once shows that the integrand $t - \ln t - 1$ arises from $\int_0^t (y_1 - 1)/y_1 \mathrm{d}s$, and the factor $\sqrt{2}$ is the normalisation constant that makes $u(y) \to 0$ as $y \to 1^-$ with unit slope. It is not a free parameter; it is fixed by the ODE boundary condition at $y = 1$. The same $\sqrt{2}$ then appears in the change of variables below and hence in the denominator of c.

The γ integral. Changing variable $u \mapsto y$ in $\int_0^\infty y_1(u)\,\mathrm{d}u$: use $\mathrm{d}u/\mathrm{d}y = -\sqrt{2}\sqrt{y - \ln y - 1}/y$, so

$$\begin{aligned}\int_0^\infty y_1(u)\,\mathrm{d}u &= \int_0^1 y\cdot\frac{\sqrt{2}\sqrt{y - \ln y - 1}}{y}\,\mathrm{d}y\\ &= \sqrt{2}\int_0^1 \sqrt{y - \ln y - 1}\,\mathrm{d}y.\end{aligned}$$

Defining γ as the integral without the $\sqrt{2}$:

$$\boxed{\gamma = \int_0^1 \sqrt{y - \ln y - 1}\,\mathrm{d}y = 0.5482228893\ldots} \tag{18}$$

The $\sqrt{2}$ from the change of variables and the $\sqrt{2}$ from the normalisation (17) combine to give the $\sqrt{2}$ in the denominator of c. This is the *one non-algebraic input* to the coefficient c; it is a one-line `scipy.quad` call.

9 Assembly: $c = X^{(2)}/(9\sqrt{2}\,\gamma)$

From LDWC eqs. (IV.16)–(IV.17), collecting the ingredients:

$$\zeta = \frac{\varepsilon}{3} + c\,\varepsilon^2 + \mathcal{O}(\varepsilon^3), \tag{19}$$

$$c = \frac{X^{(2)}}{9\sqrt{2}\,\gamma} = \frac{1}{9\sqrt{2}\times 0.54822\ldots} = 0.143313\ldots$$

matching the LDWC published value $c = 0.14331$ (relative error 2×10^{-4}).

Remark 6 (Where $9 = 3^2$ comes from)**.** Integrating the 2-loop FRG flow equation (LDWC eq. III.42) over $u \in [0, \infty)$ and evaluating at the one-loop fixed point $\Delta^* = (\varepsilon/3)\, y_1(u) + O(\varepsilon^2)$ gives (LDWC eq. IV.2):

$$-3\,c_2\,\varepsilon^2 \cdot \tfrac{\varepsilon}{3}\sqrt{2}\gamma = -X^{(\alpha)}\left(\tfrac{\varepsilon}{3}\right)^3, \tag{20}$$

where $\varepsilon/3 = \int_0^\infty \Delta^*\mathrm{d}u/\sqrt{2}\gamma$ and $(\varepsilon/3)^3 = -\Delta'(0^+)^3$. yielding $c_2 = X^{(\alpha)}/(27\sqrt{2}\gamma)$ in LDWC's convention $\zeta = \varepsilon/3 + c_2\varepsilon^2$. With our convention $\zeta = (\varepsilon/3)(1 + c\varepsilon)$, i.e. $c = 3c_2$:

$$c = \frac{X^{(\alpha)}}{9\sqrt{2}\,\gamma}.$$

The factor $9 = 3^2$ is ζ_1^{-2} where $\zeta_1 = 1/3$ is the one-loop roughness exponent. Both factors of 3 come from the one-loop fixed-point amplitude $\Delta'(0^+) = -\varepsilon/3$: one from $\int \Delta^*\,\mathrm{d}u = (\varepsilon/3)\sqrt{2}\gamma$ and two from $\Delta'(0^+)^3 = (-\varepsilon/3)^3$.

Every row above the residual line is algebraic. The residual line has $N_{\text{res}} = 1$ entry.

10 Machine verification

The pipeline is implemented in the `rdft` library and the derivation chain is verified by a passing pytest class `TestLDWCoefficientRecovery` (`tests/test_parametric_analytic.py`):

1. `test_psi_sunset_is_first_symanzik` — Ψ_{sun} is degree 2 (not K, which is degree 1).
2. `test_psi_sunset_matches_ldw_denominator` — $\Psi(\alpha, 1, \beta) = \alpha + \beta + \alpha\beta$.
3. `test_J1_equals_ln2_analytically` — SymPy proves $J_1 = \ln 2$ from first principles.
4. `test_ln2_cancellation_J1_plus_J2` — algebraic cancellation $J_1 + J_2 = 0$.
5. `test_X2_equals_one` — $X^{(2)} = 1$.
6. `test_gamma_integral_ldw` — $\gamma = 0.5482\ldots$
7. `test_ldw_coefficient_recovery` — end-to-end $c = 0.14331$, relative error $< 10^{-3}$.

11 What is and is not claimed

What CFAC establishes here.

- Class-selection filters (causal contraction, bridge factorisation) discard LDWC classes B and C without invoking FRG-specific cancellations.
- The first Symanzik polynomial Ψ_{sun} is the denominator of the LDWC integral J; $\beta = 1$ is its symmetry point, explaining why LDWC's domain split works.

Ingredient	Value	Determined by
I_{tad}	$(4\pi)^{-d/2}\Gamma(1-d/2)m^{d-2}$	Gamma function
I_{bub}	$(4\pi)^{-d/2} \cdot$ (algebra)	Gamma function
$I_R(\text{sun})$ (BPHZ)	$X^{(2)} = 1$, no new integral	BPHZ + J-decomposition
J_1	$\ln 2$	SymPy integral from $\mathit{\Psi}_{\text{sun}}$
J_2	$-\ln 2$	$\beta \to 1/\beta$ symmetry of $\mathit{\Psi}_{\text{sun}}$
$J_3\|_{\text{fin}}$	1 (rational)	MS renormalisation of UV pole
$X^{(2)}$	1	$J_1 + J_2 + J_3\|_{\text{fin}}$
γ	$0.5482\ldots$	**[residual]** `scipy.quad`
c	0.14331	$X^{(2)}/(9\sqrt{2}\gamma)$

Table 3: Complexity account: every row above the residual line is algebraic. The residual list has $N_{\text{res}} = 1$ entry.

- $J_1 = \ln 2$ is derived from multilinearity of $\mathit{\Psi}_{\text{sun}}$.
- $J_2 = -\ln 2$ is derived from the $\beta \leftrightarrow 1/\beta$ symmetry of $\mathit{\Psi}_{\text{sun}}$.
- $J_3|_{\text{finite}} = 1$ is derived from the Schwinger asymptotic of $\mathit{\Psi}_{\text{sun}}$ (Section 7).
- The prefactor $9 = \zeta_1^{-2}$ is derived from the integrated one-loop FRG equation (Section 9).
- The residual count $N_{\text{res}} = 1$: exactly one 1D integral (γ) is needed; everything else is algebra.

What is still taken from LDWC.

- The FRG Feynman rules (propagators, vertices) for the elastic-manifold theory. This is physics input for any field-theory calculation.
- The enumeration of raw 2-loop diagrams before filtering. Class A/B/C classification as drawn in LDWC figs. III.2–III.5.
- The one-loop fixed-point function $y_1(u)$ and its normalisation $y_1'(0^+) = -1$, which give $\Delta'(0^+) = -\varepsilon/3$ (LDWC eqs. IV.5–IV.12). The ODE itself *is* rederived here; only LDWC's specific scaling convention is adopted.

12 Open directions

Short-range case: $N_{\text{res}} = 1$, $X^{(1)} = \ln 2$. The period integral of $1/\mathit{\Psi}_{\text{sun}}$ over the standard simplex is $\ln 2$ — a single transcendental, still in the residual list but now topological. Computing it from $\mathit{\Psi}_{\text{sun}}$ directly (without quoting LDWC) is a one-line extension of Proposition 2.

Three loops. At three loops the "Mercedes" diagram has no cut vertex and a more complex $\mathit{\Psi}$; the "banana" factorises into three bubbles. The pipeline extends straightforwardly; N_{res} is expected to grow by at most one per genuine new ODE mode.

General bound on N_{res}. For an L-loop FRG calculation with r independent one-loop fixed-point modes, we conjecture $N_{\text{res}} \le r$. At two loops $r = 1$ and $N_{\text{res}} = 1$ as shown. A proof would require classifying which diagram periods are periods of lower-loop diagrams (Euler characteristic argument).

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